

System administration is not a child's play. Be prepared to counter lots and lots of commands through this section. Most of the commands are restricted to a normal user and so you have to don the root's cap. With this power comes great responsibility!

This chapter will take you from booting your machine (also called bootstrapping) and then on to various runlevels and services of the OS. Commands are introduced mid way; these commands deal with system monitoring and certain level of tweaking. Upgrading software is discussed in section 9.4; this primarily deals with yum and apt-get. We will wind this chapter with Kernel Compilation—both Rebuilding and New Compilation.

9.1 Boot Loader And Tweaking Boot Process

The booting process of an operating system is very important because it differentiates between a functioning and a malfunctioning system. Many home users tend to use Linux as the secondary operating system and dual-boot it with Windows as their primary operating system. The use of Windows as a primary OS is because of the lack of information on Linux, and also because many games and software are not available for Linux; for example: Microsoft Office.

As the administrator of your system you have the power to make changes to the boot process. You can change the services that start at boot, alter the settings of the boot loader, as well as use other powerful tools such as the single-user mode.

Before we introduce you to the power of 'root', let us tell you a little about how Linux handles booting in general. We will also consider the dual-boot part, as it will also include the MBR section in detail.

Technically, one can create four primary partitions of which one is extended to a virtual area called 'Extended Partition' in order to allow creation of more partitions. Under this Extended partition, a user is allowed to create up to 64 'Logical Drives'.